

Digital Electronics

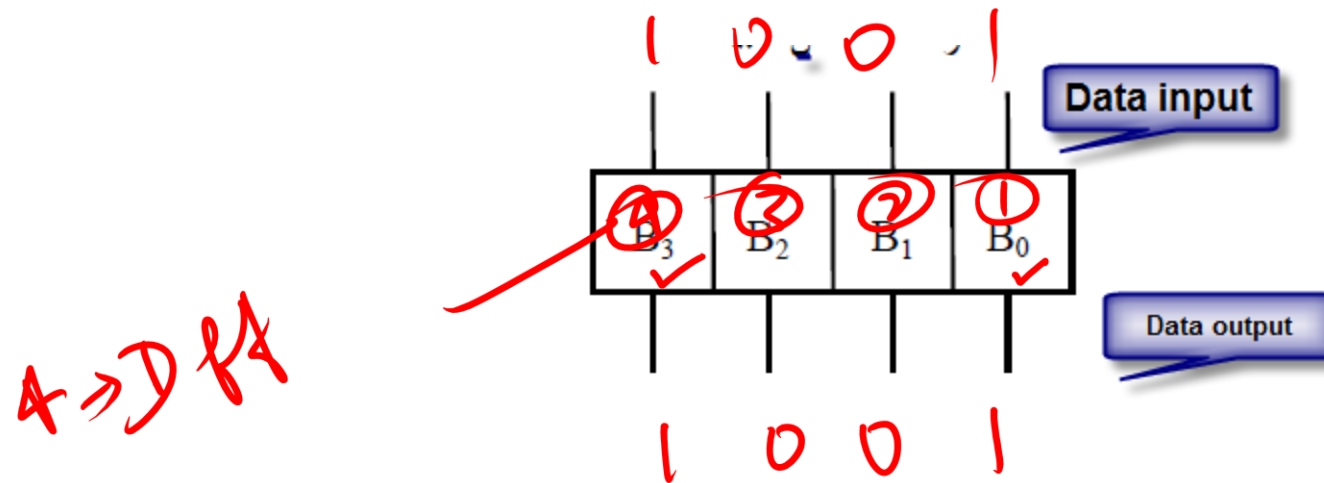
III Semester

Registers & Counter

- Circuits that include flip-flops are usually classified by the function they perform
 - Registers
 - Counters
- **Register** is a group of flip-flops.
 - Each flip-flop is capable of storing one bit of information.
 - An n-bit register consists of a group of n flip-flops.
 - Register is a group of binary cells suitable for holding binary information.
- A **counter** is essentially a register that goes through a predetermined sequence of states.

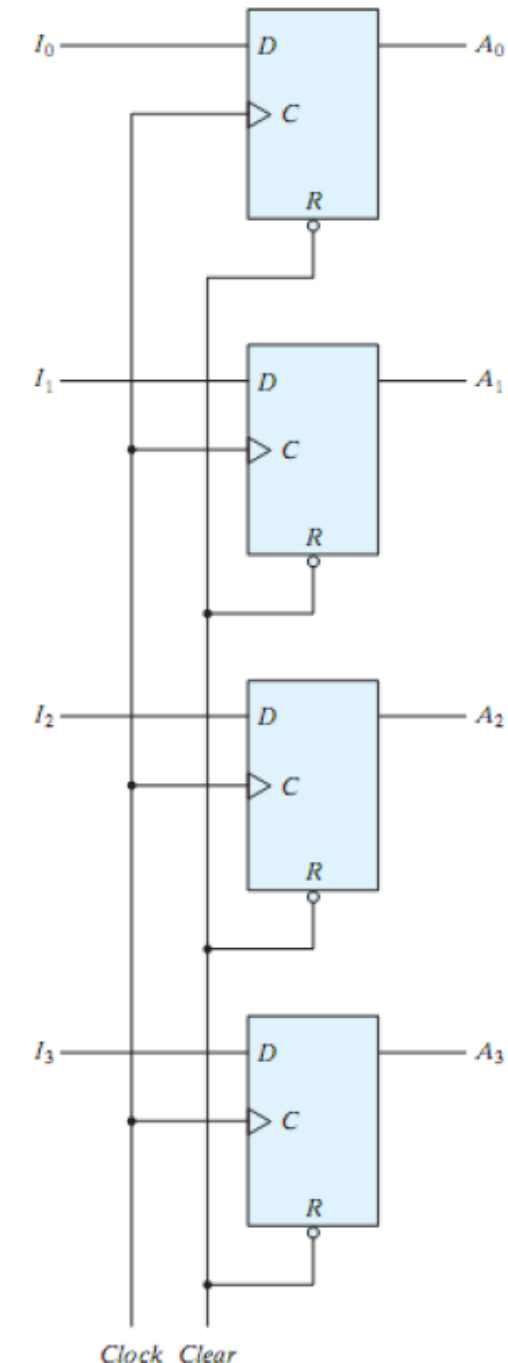
Registers

- Registers is storing Element example 4-bit Register



4-Bit Register

- A simple 4-bit register can be made with 4 D-FF
 - Data is loaded in parallel
- Common Clock
 - At each positive-edge, 4 bits are loaded in parallel
 - Previous data is overwritten
 - Entering data is called **loading**
- Common Clear
 - Asynchronous active-low clear
 - When Clear = 0, all FFs are cleared; i.e. 0 is stored.

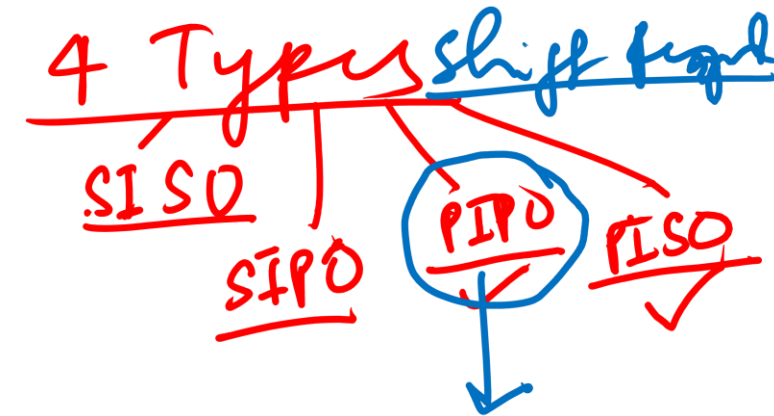
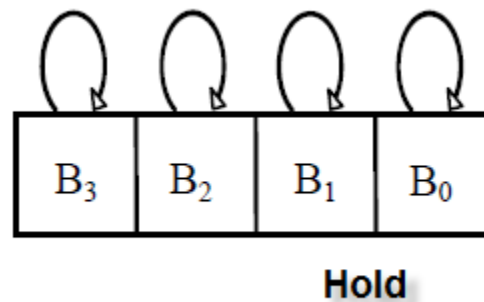
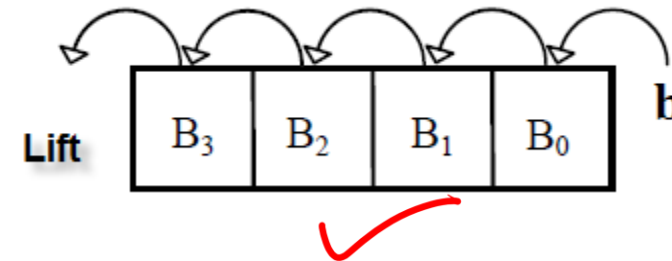
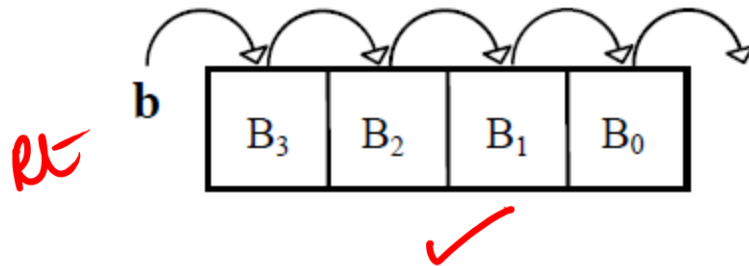


SHIFT REGISTERS

- In digital circuits a **shift register** is a group of flip flops set up in a linear fashion which have their inputs and outputs connected together in such a way that the data are shifted down the line when the circuit is activated.
- The simplest shift register is one that uses only flip-flops.

Shift Registers

- Shift Register Shift information 1bit Right or Left



Registers

1 bit storage - 1 FF

n bit Register $\equiv$ n flipflops

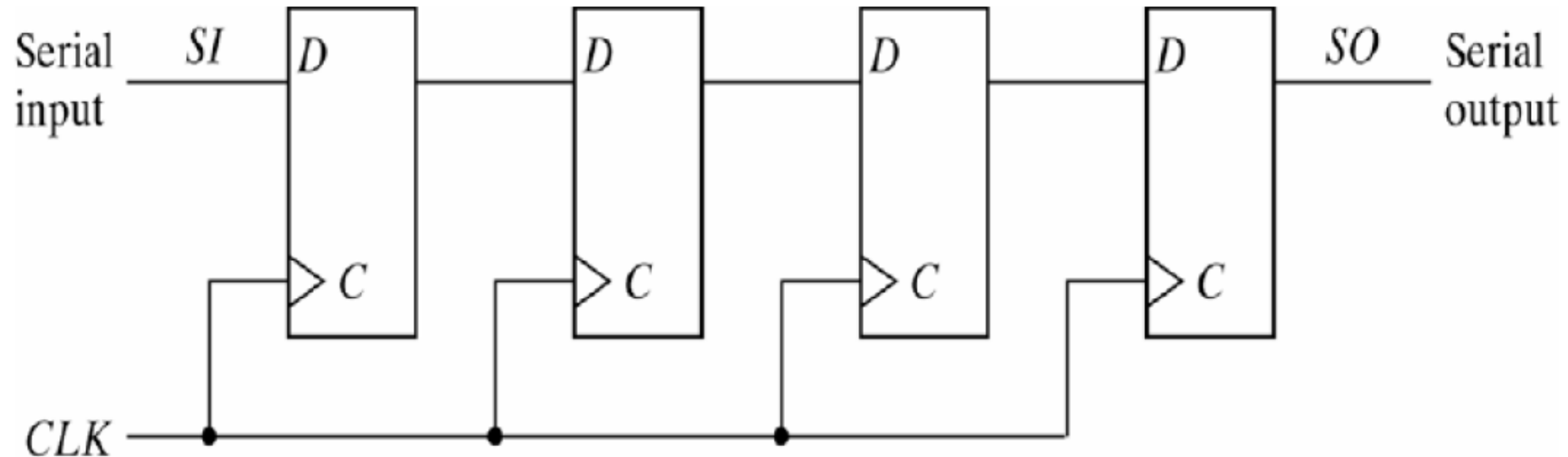
8 bit " = 8 ff (^{the edge} D type FF)

= Storage of data

→ shifting of data $\Rightarrow$ shift register

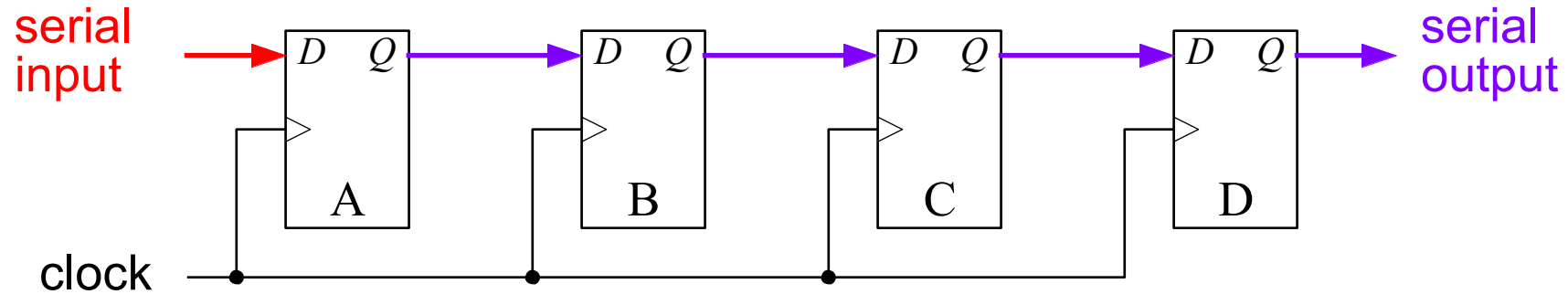
Shift Registers

- Capable of shifting its binary information in one or both directions



A Simple Shift Register

A basic shift register is simply a chain of *D* flip-flops with a common clock.



Each flip-flop transfers its *D* input to its *Q* output at a clock transition.

- The effect is to transfer data along the register, one flip-flop per clock cycle. This type of register is called a serial input-serial output (SISO).

- Figure shows a four-bit shift register that is used to shift its contents one bit-position to the right.
- The data bits are loaded into the shift register in a serial fashion using the In input.
- The contents of each flip-flop are transferred to the next flip-flop at each positive edge of the clock.

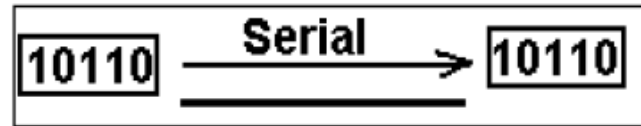
Shift Register Applications

Shift Registers are an important Flip-Flop configuration with a wide range of applications, including:

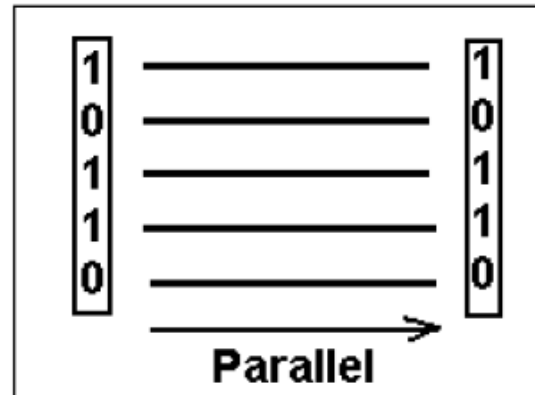
- Computer and Data Communications
- Serial and Parallel Communications
- Multi-bit number storage
- Sequencing
- Basic arithmetic such as scaling (a serial shift to the left or right will change the value of a binary number a power of 2)
- Logical operations

Parallel versus Serial

- Serial communications: provides a binary number as a sequence of binary digits, one after another, through one data line.

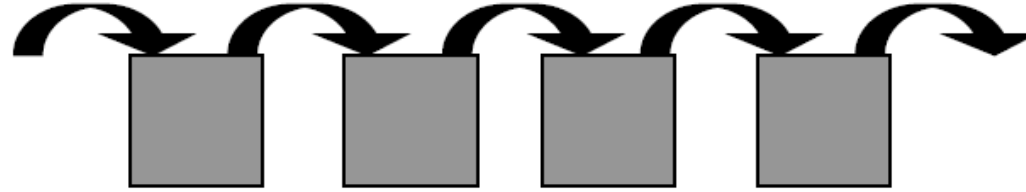


- Parallel communications: provides a binary number as binary digits through multiple data lines at the same time.

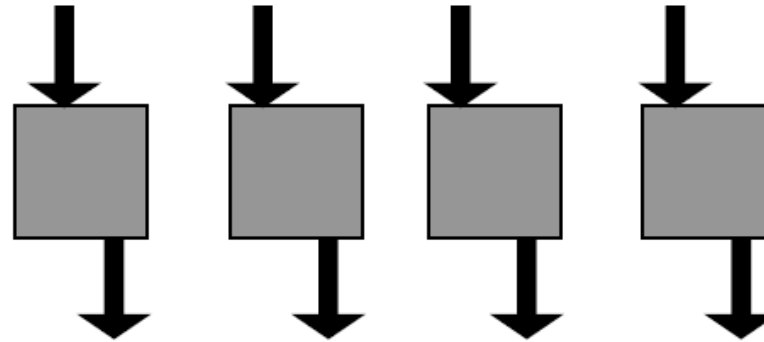


Shift Registers

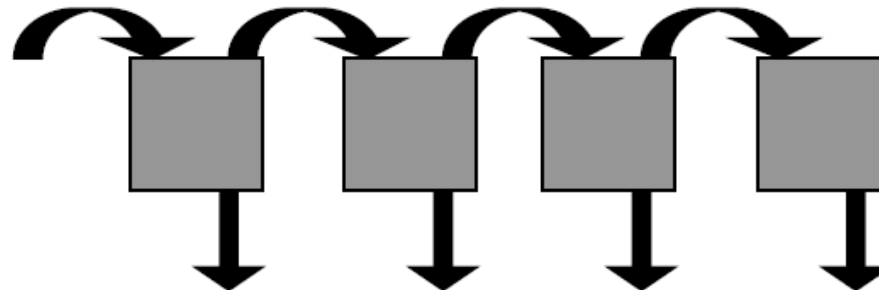
- Shift Registers are devices that store and move data bits in serial (to the left or the right),



- ..or in parallel,

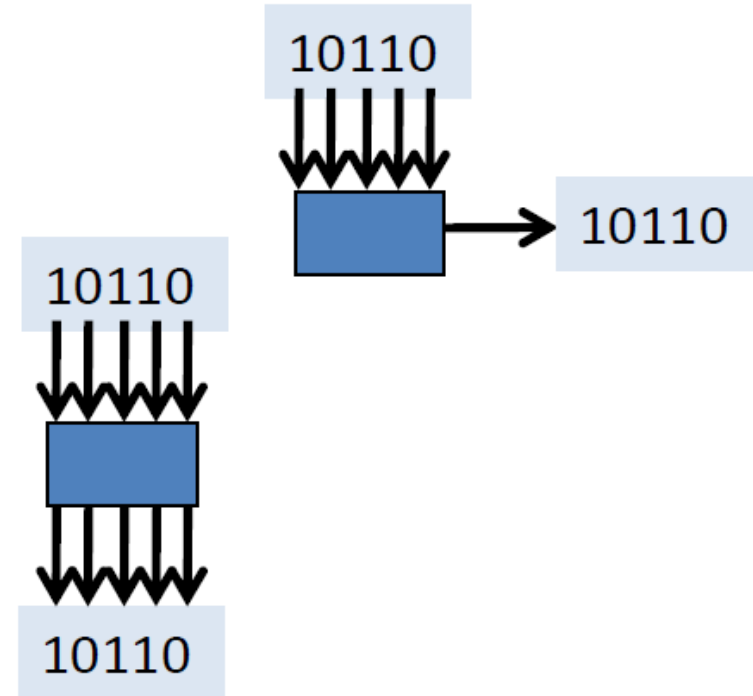
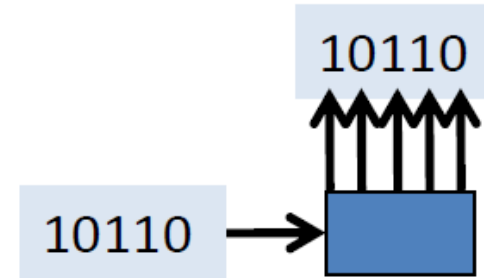


- ..or a combination of serial and parallel.



Combinations of Data Transfer Methods

- SISO: Serial In, Serial Out
- SIPO: Serial In, Parallel Out
- PISO: Parallel In, Serial Out
- PIPO: Parallel In, Parallel Out

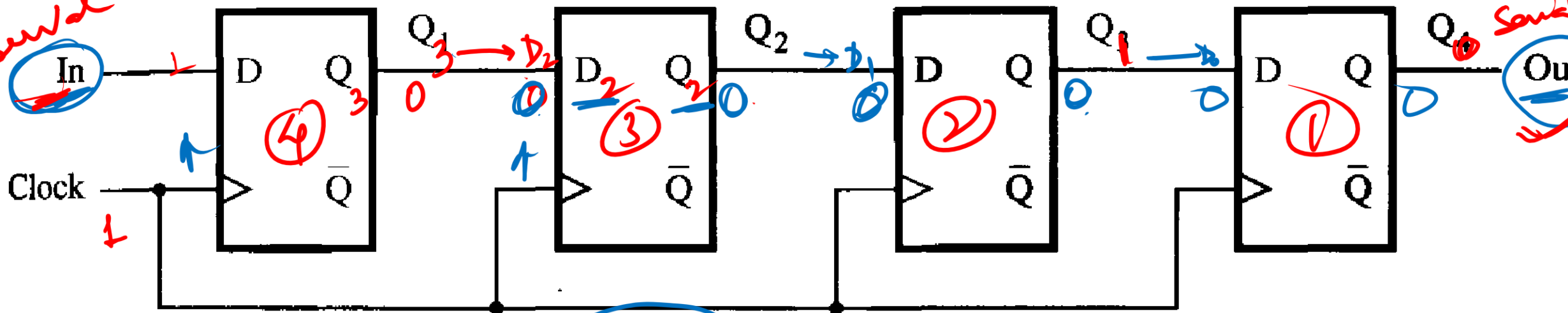


SISO

Serial In

Serial Out

~~0 0 0 1~~ LSB



7 clk pulses

7th
5th
4th
3rd
2nd
1st

Serial IN
~~D₃ D₂ D₁ D₀~~
(LSB)

Q₃ Q₂ Q₁ Q₀

Serial out

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

RA Shift Register

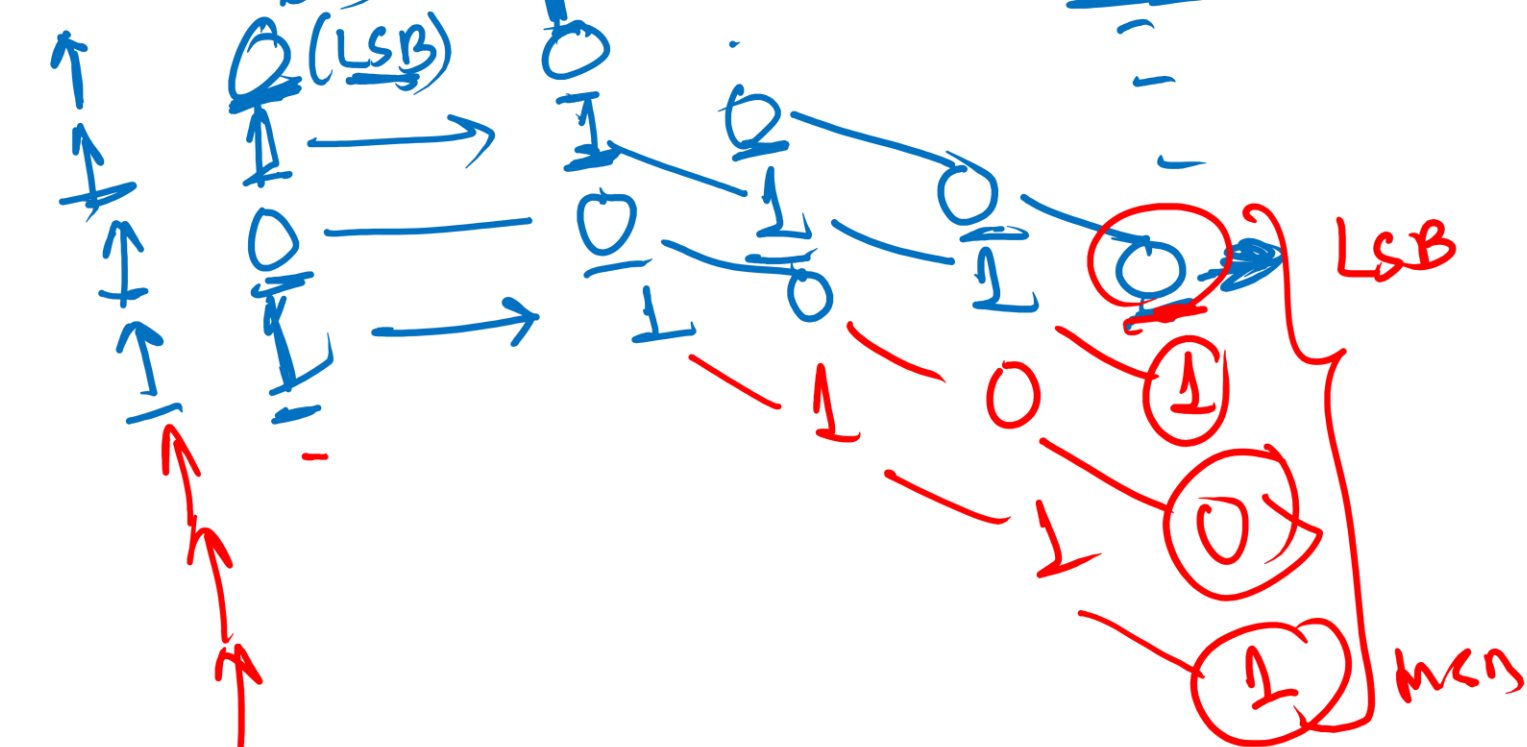
1 (LSB)

SISO

Data in
Ds

clk

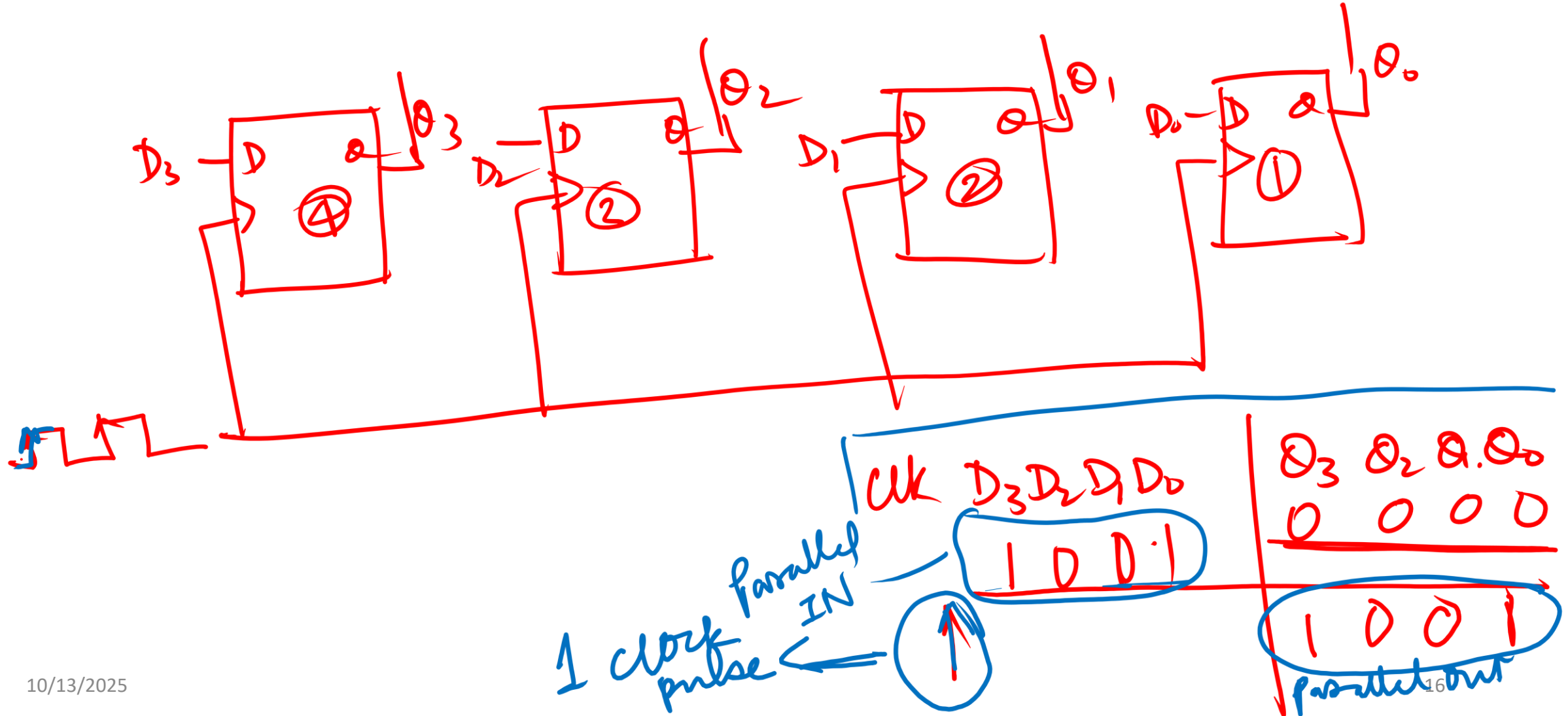
Q_3 Q_2 Q_1 Q_0/Q_s



4 bits
4 period
7

Register - 4 bits

PIPO → n bit Register
1 clk pulse



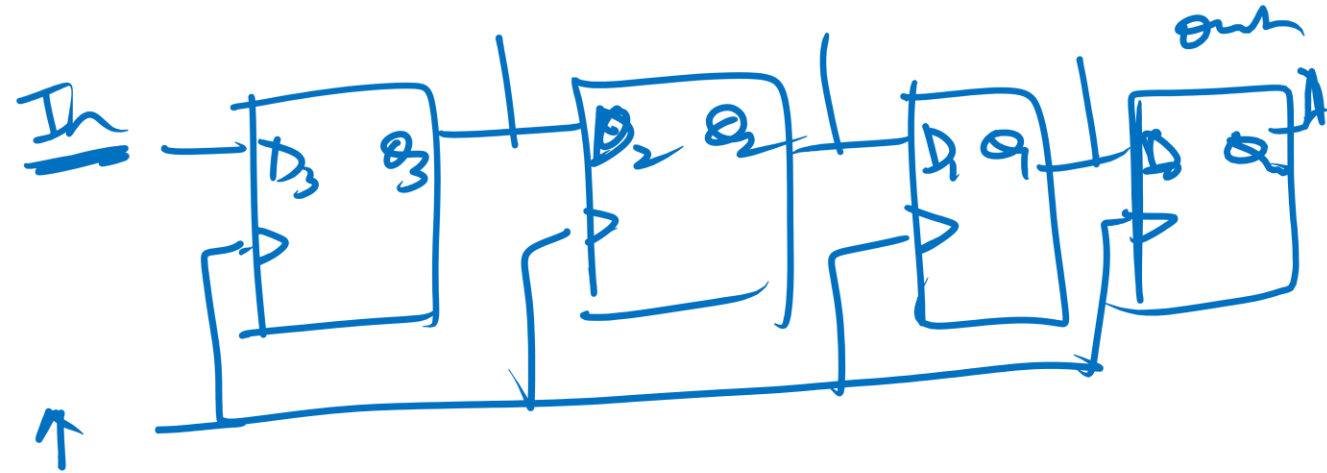
- An illustration of the transfer is shown below, which shows what happens when the signal values at In during eight consecutive clock cycles are 1,0,1,1,1,0,0, and 0, assuming that the initial state of all flip-flops is 0.

Serial out

	In	Q_3	Q_2	Q_1	$Q_0 = \text{Out}$
t_0	① 1	0	0	0	0
t_1	② 0	1	0	0	0
t_2	③ 1	0	1	0	0
t_3	④ 1	1	0	1	0
t_4	⑤ 1	1	1	0	1
t_5	⑥ 0	1	1	1	0
t_6	⑦ 0	0	1	1	1
t_7	⑧ 0	0	0	1	1

Serial out

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ Shift right



- To implement a shift register, it is necessary to use either edge-triggered or master-slave flip-flops. The level-sensitive gated latches are not suitable, because a change in the value of In would propagate through more than one latch during the time when the clock is equal to 1.

Registers

$$\frac{4 \text{ ff's}}{n \text{ ff's.}}$$

SISO

SIPO

PIPO

PISO

$$4 \times 2 - 1 =$$

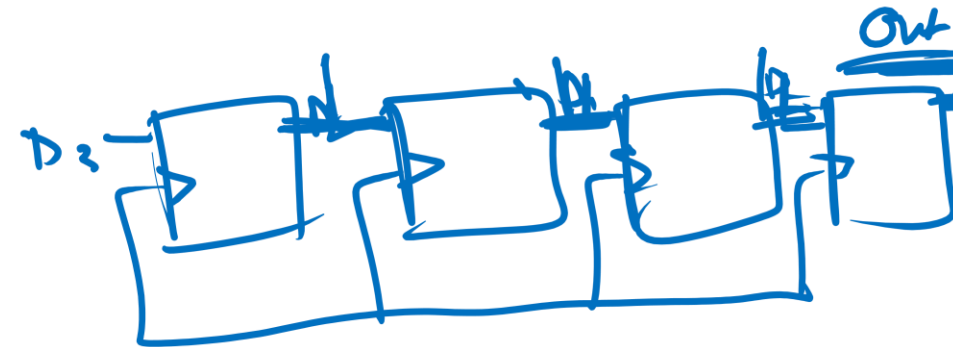
$$\frac{7 \text{ clk}}{8}$$

$$n + n = \underline{2n - 1}$$

$$\frac{4 \text{ clk}}{n}$$

$$\begin{matrix} 1 \text{ clk} \\ 1 \text{ clk} \end{matrix}$$

$$\begin{matrix} 4 \text{ clk} \\ n \text{ clk} \end{matrix}$$



4 SISO unidirectional shift register 1011

$D_s = \underline{1}, \underline{1}, \underline{0}, \underline{1}, 0, 1, 0$ - Take data through ff's &

If data is read through $Q_s \rightarrow$ How many clk pulses are required

"01011" with LSB first

11:25

clk	D_s	Q_3	Q_2	Q_1	Q_0/Q_s
0	1	1	0	1	1
1	1	1	1	0	1
2	0	1	1	1	0
3	1	1	1	1	1
4	0	1	0	1	1
5	1	0	1	0	1
6	0	1	0	1	0
7	0	0	1	0	1

After 4 clk pulses

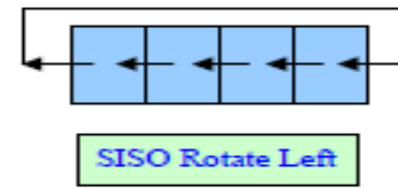
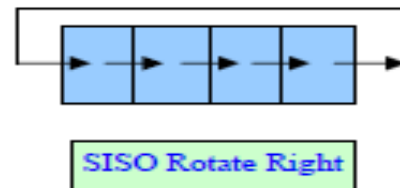
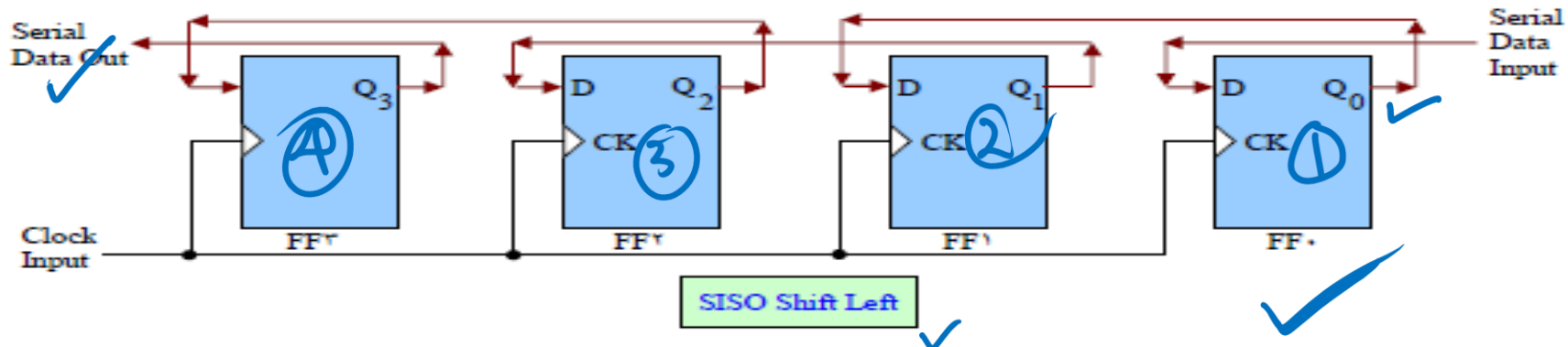
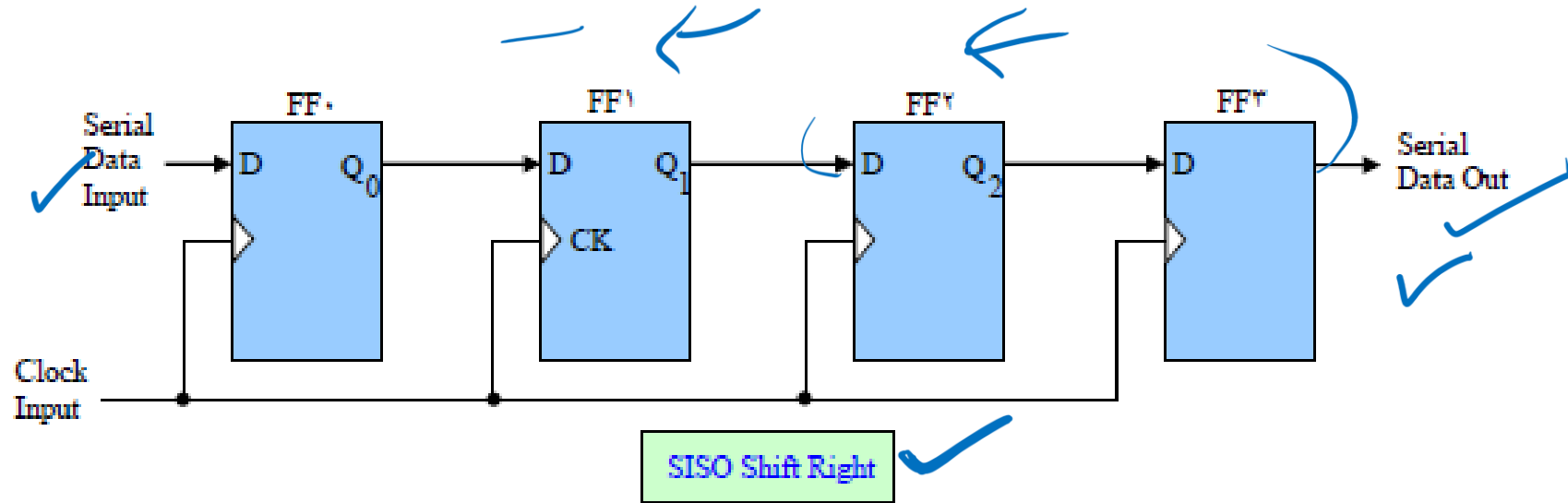
5th clk pulse

6th clk pulse

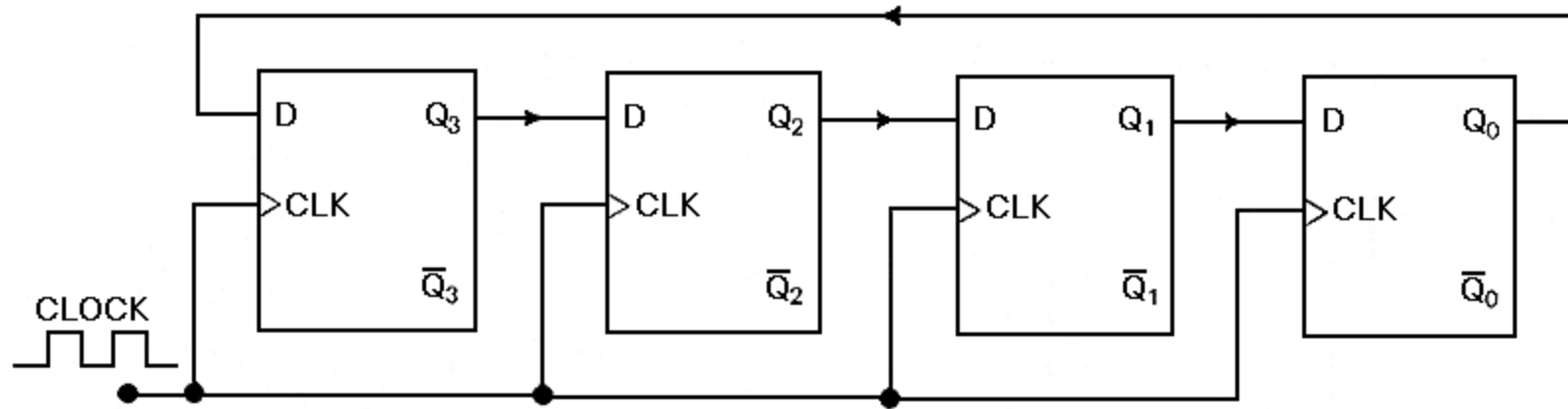
7th clk pulse

$2n-1$
7 bits as I/P
 $\frac{4 \text{ bits} + 6 \text{ clk}}{10 \text{ clk}}$

Serial in Serial out using D FF

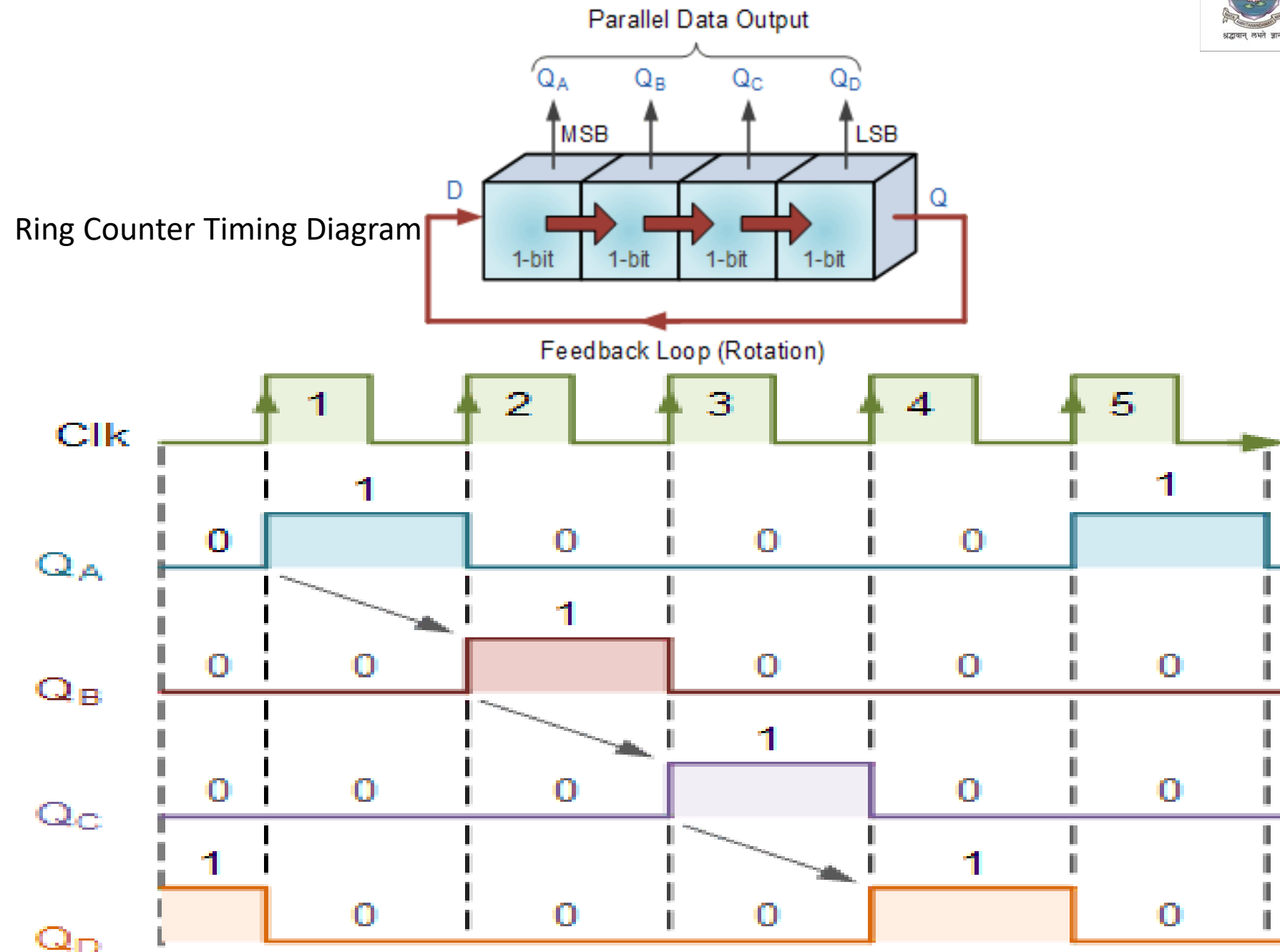


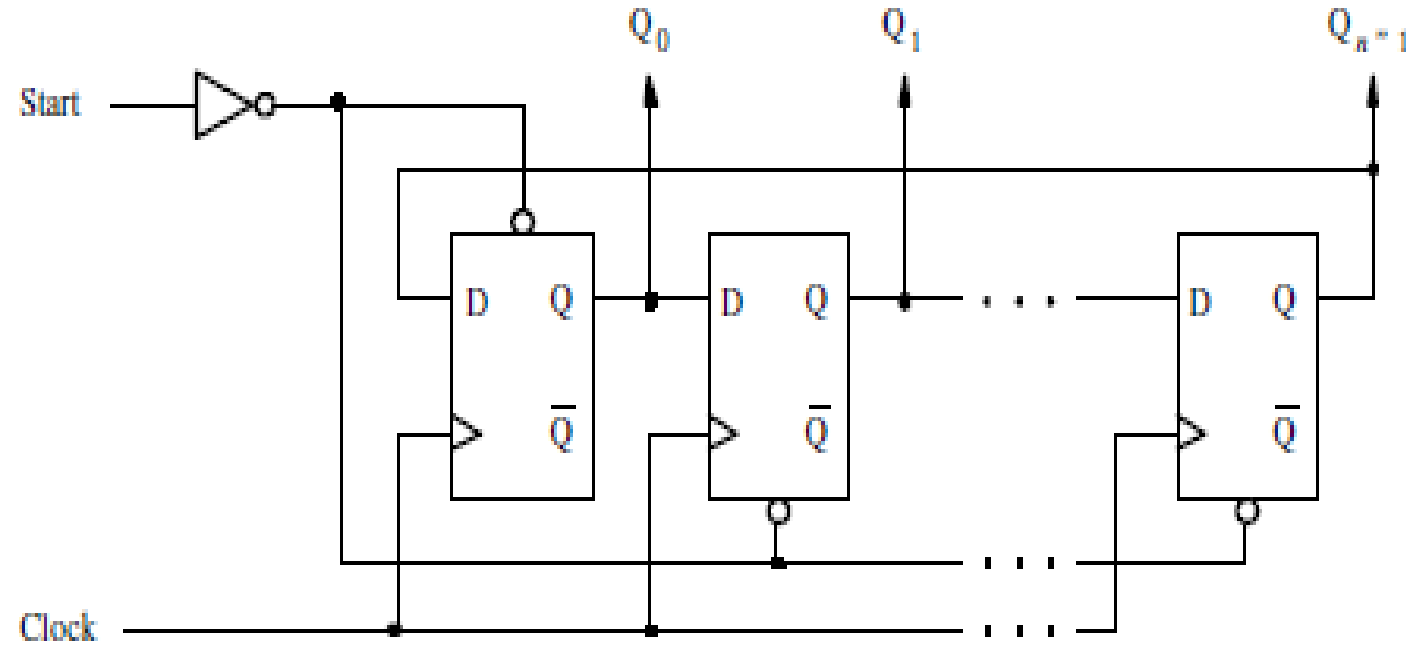
❖ Ring Counter



Ring counters are implemented using shift registers. It is essentially a circulating shift register connected so that the last flip-flop shifts its value into the first flip-flop. There is usually only a single 1 circulating in the register, as long as clock pulses are applied.

(Starts 1000- $\rightarrow$ 0100- $\rightarrow$ 0010- $\rightarrow$ 0001 repeat)

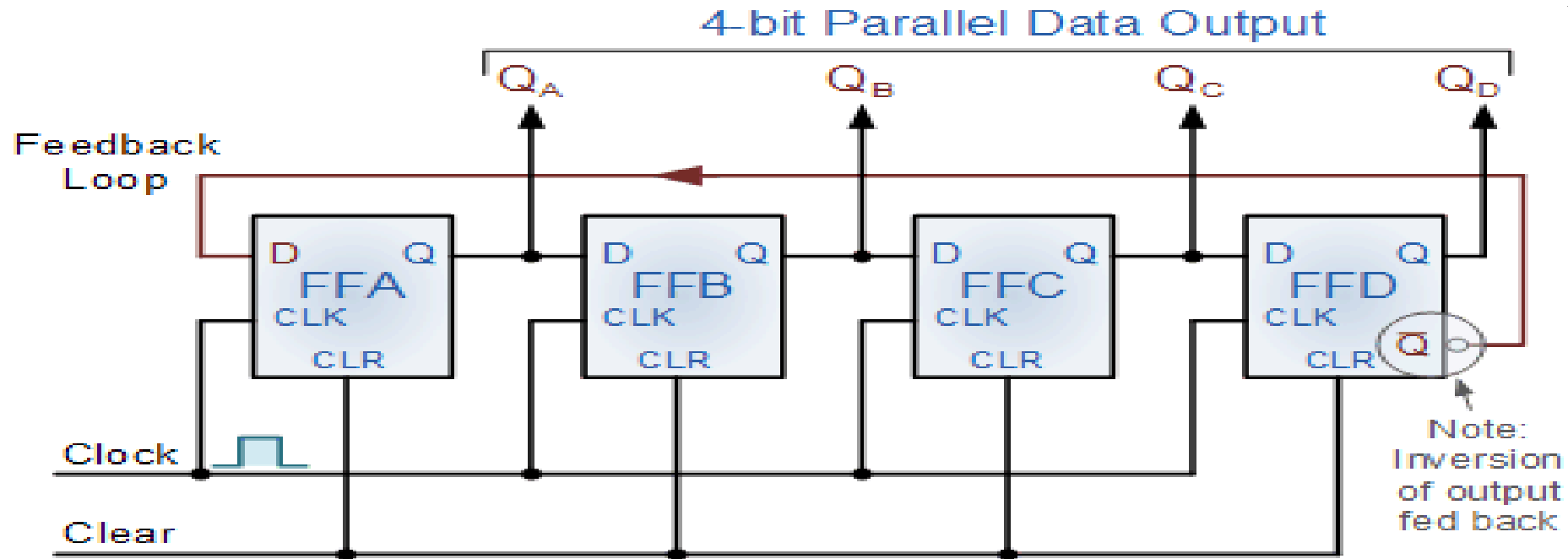




(a) An n -bit ring counter

Start control signal, which presets the left-most flip-flop to 1 and clears the others to 0.

❖ Johnson Counter



The Johnson counter, also known as the twisted-ring counter, is exactly the same as the ring counter except that the inverted output of the last flip-flop is connected to the input of the first flip-flop.

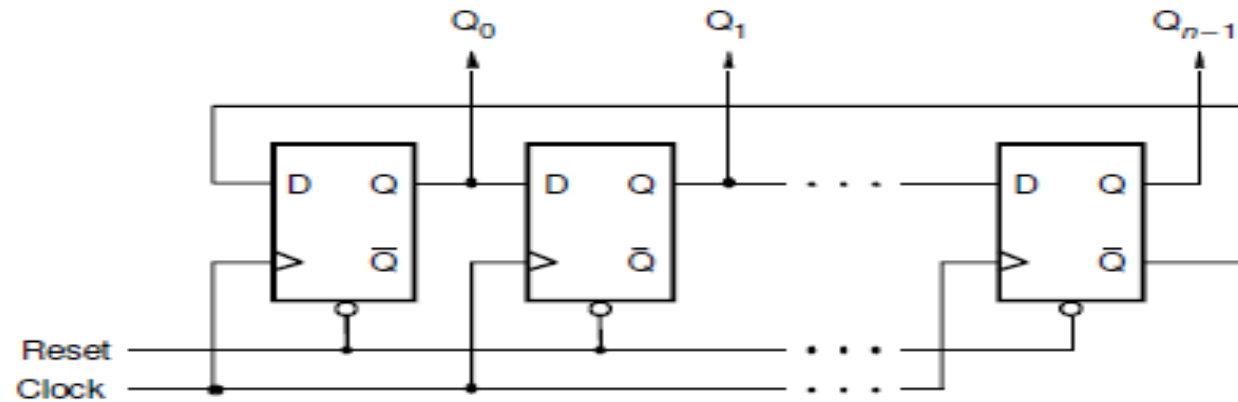
Let's say, starts from 000, 100, 110, 111, 011 and 001, and the sequence is repeated so long as there is input pulse.

**Truth Table for a 4-bit Johnson
/Twisted Ring Counter**

Clock Pulse No	FFA	FFB	FFC	FFD
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1



Johnson Counter



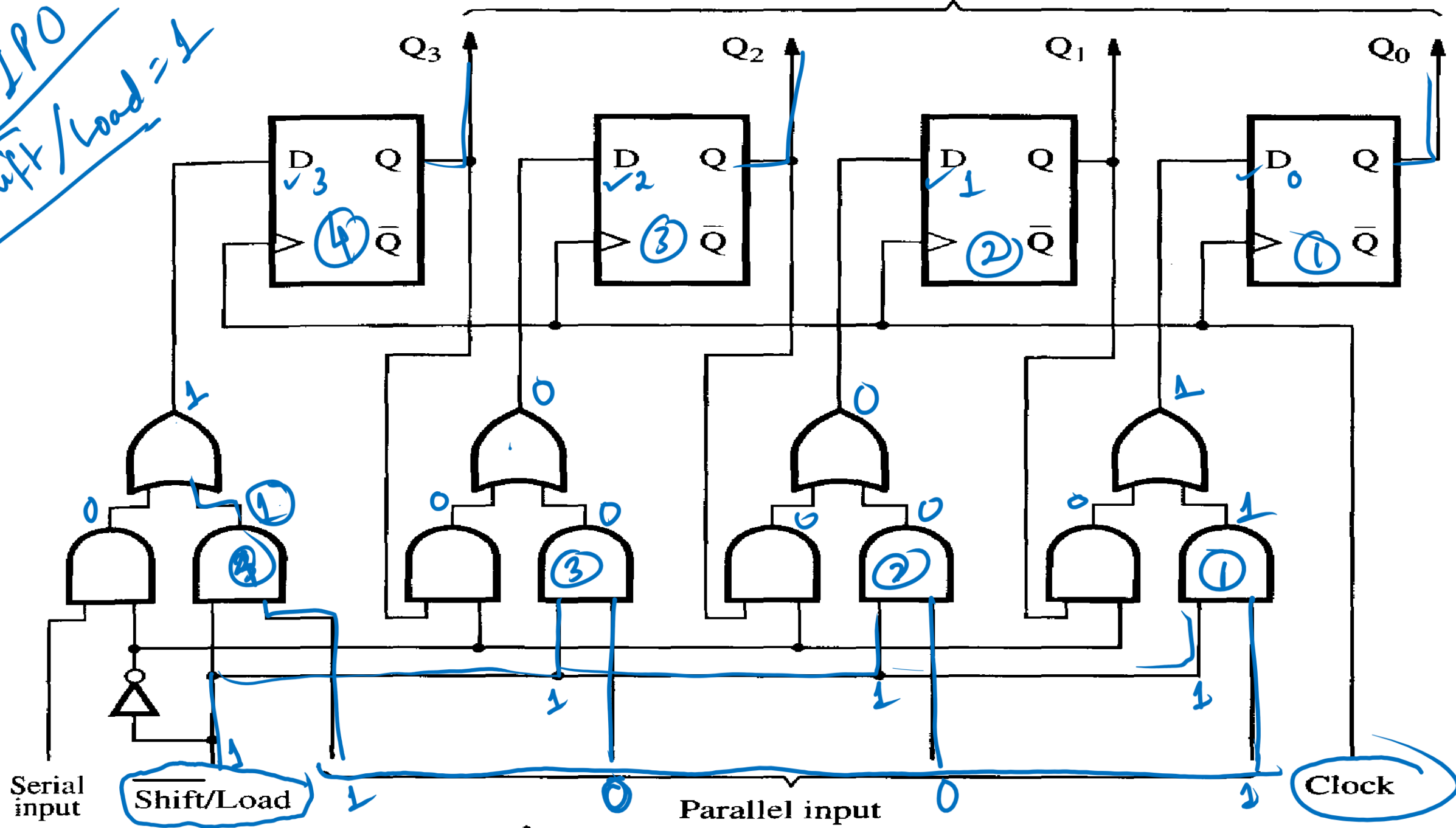
To initialize the operation of the Johnson counter, it is necessary to reset all flip-flops, as shown in the figure. Observe that neither the Johnson nor the ring counter will generate the desired counting sequence if not initialized properly.

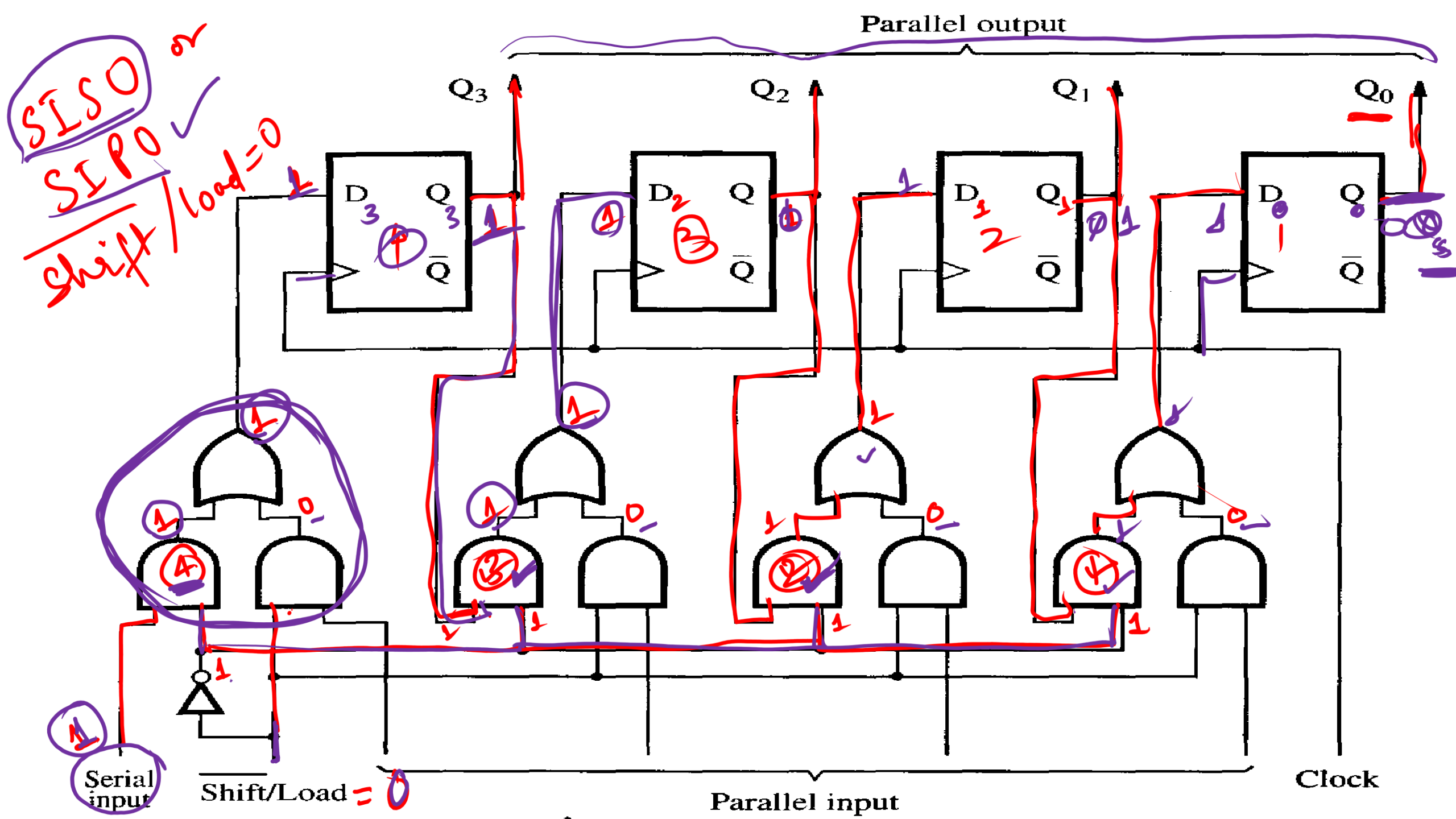
PARALLEL-ACCESS SHIFT REGISTER

In computer systems it is often necessary to transfer n -bit data items. This may be done by transmitting all bits at once using n separate wires, in which case we say that the transfer is performed in *parallel*. But it is also possible to transfer all bits using a single wire, by performing the transfer one bit at a time, in n consecutive clock cycles. We refer to this scheme as *serial* transfer. To transfer an n -bit data item serially, we can use a shift register that can be loaded with all n bits in parallel (in one clock cycle). Then during the next n clock cycles, the contents of the register can be shifted out for serial transfer. The reverse operation is also needed. If bits are received serially, then after n clock cycles the contents of the register can be accessed in parallel as an n -bit item.

Parallel output

↑ IPO
Shift/Load = 1





Parallel output

